$$K^{(a)} = \frac{1}{\sqrt{2}} \overbrace{\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}}^{H \otimes I} \cdot \overbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}}^{|0\rangle \otimes I} = \overbrace{\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}}^{K^{(a)}} \begin{matrix} K_0 \\ \\ K_1 \\ \\ \end{matrix}$$

slices $I/\sqrt{2}$, $I/\sqrt{2}$

$$\overbrace{K^{(b)} = \mathrm{CX}\, K^{(a)}}^{} \qquad \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{matrix} K_0 \\ \\ K_1 \\ \\ \end{matrix}$$

slices $I/\sqrt{2}$, $X/\sqrt{2}$